

Quantum Field Theory - Homework 1

9,1/10

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Problem 1. 1,8/2

Let $J_{\mu\nu}$ be the generators of the Lorentz group.

(a) Show that the Lie algebra of the Lorentz group is

$$i[J^{\alpha\beta}, J^{\rho\sigma}] = g^{\sigma\beta} J^{\rho\alpha} + g^{\rho\alpha} J^{\sigma\beta} + g^{\rho\beta} J^{\sigma\alpha} + g^{\sigma\alpha} J^{\beta\rho}.$$

(b) Obtain the irreducible representations of the Lorentz group.

Solution**Part (a)**

To obtain the Lie algebra of the Lorentz group we can first obtain the generators in the vector representation, which are given by

$$\begin{aligned} \delta x^\mu &= x'^\mu - x^\mu = \delta\omega^\mu{}_\nu x^\nu \\ &= -\frac{1}{2}(\delta\omega_{\mu\nu} - \delta\omega_{\nu\mu})x^\nu \\ &= -\frac{1}{2}\delta\omega_{\alpha\beta}(g^{\alpha\mu}g^{\beta\nu} - g^{\beta\mu}g^{\alpha\nu})x_\nu \\ &= \frac{i}{2}\delta\omega_{\alpha\beta} [i(g^{\alpha\mu}x^\beta - g^{\beta\mu}x^\alpha)] \end{aligned} \quad (1)$$

core full with index placement

using the fact that $\partial_\mu x^\nu = \delta_\mu^\nu$:

$$\delta x^\mu = \frac{i}{2}\delta\omega_{\alpha\beta} \underbrace{[i(x^\beta\partial^\alpha - x^\alpha\partial^\beta)]}_{J^{\alpha\beta}} x^\mu = \frac{i}{2}\delta\omega_{\alpha\beta} J^{\alpha\beta} x^\mu \quad (2)$$

so that the generators of the Lorentz group in the vector representation are given by

$$J_{\mu\nu} = i(x_\mu\partial_\nu - x_\nu\partial_\mu) \quad (3)$$

proceeding in this way isn't the most general, since we are choosing a specific representation of the generators, but we can use the fact that the Lie algebra is representation independent, so that the commutation relations between the generators will be the same in any representation.

From that, we can compute the commutator directly:

$$\begin{aligned} [J^{\alpha\beta}, J^{\rho\sigma}] &= i^2 [(x^\alpha\partial^\beta - x^\beta\partial^\alpha), (x^\rho\partial^\sigma - x^\sigma\partial^\rho)] \\ &= (x^\alpha\partial^\beta - x^\beta\partial^\alpha)(x^\rho\partial^\sigma - x^\sigma\partial^\rho) - (x^\rho\partial^\sigma - x^\sigma\partial^\rho)(x^\alpha\partial^\beta - x^\beta\partial^\alpha) \\ &= x^\alpha(\partial^\beta x^\rho)\partial^\sigma + x^\alpha x^\rho\partial^\beta\partial^\sigma - x^\alpha(\partial^\beta x^\sigma)\partial^\rho - x^\alpha x^\sigma\partial^\beta\partial^\rho - x^\beta(\partial^\alpha x^\rho)\partial^\sigma - x^\beta x^\rho\partial^\alpha\partial^\sigma \\ &\quad + x^\beta(\partial^\alpha x^\sigma)\partial^\rho + x^\beta x^\sigma\partial^\alpha\partial^\rho - x^\rho(\partial^\sigma x^\alpha)\partial^\beta - x^\rho x^\alpha\partial^\sigma\partial^\beta + x^\rho(\partial^\sigma x^\beta)\partial^\alpha + x^\rho x^\beta\partial^\sigma\partial^\alpha \\ &\quad + x^\sigma(\partial^\rho x^\alpha)\partial^\beta + x^\sigma x^\alpha\partial^\rho\partial^\beta - x^\sigma(\partial^\rho x^\beta)\partial^\alpha - x^\sigma x^\beta\partial^\rho\partial^\alpha \\ &= g^{\beta\rho}x^\alpha\partial^\sigma - g^{\beta\sigma}x^\alpha\partial^\rho - g^{\alpha\rho}x^\beta\partial^\sigma + g^{\alpha\sigma}x^\beta\partial^\rho - g^{\sigma\alpha}x^\rho\partial^\beta + g^{\sigma\beta}x^\rho\partial^\alpha + g^{\rho\alpha}x^\sigma\partial^\beta - g^{\rho\beta}x^\sigma\partial^\alpha \\ &= g^{\alpha\rho}(-x^\beta\partial^\sigma + x^\sigma\partial^\beta) + g^{\alpha\sigma}(x^\beta\partial^\rho - x^\rho\partial^\beta) + g^{\beta\rho}(x^\alpha\partial^\sigma - x^\sigma\partial^\alpha) + g^{\beta\sigma}(-x^\alpha\partial^\rho + x^\rho\partial^\alpha) \\ &= ig^{\alpha\rho}[i(x^\beta\partial^\sigma - x^\sigma\partial^\beta)] - ig^{\alpha\sigma}[i(x^\beta\partial^\rho - x^\rho\partial^\beta)] - ig^{\beta\rho}[i(x^\alpha\partial^\sigma - x^\sigma\partial^\alpha)] + ig^{\beta\sigma}[i(x^\alpha\partial^\rho - x^\rho\partial^\alpha)] \\ &= i(g^{\alpha\rho}J^{\beta\sigma} - g^{\alpha\sigma}J^{\beta\rho} - g^{\beta\rho}J^{\alpha\sigma} + g^{\beta\sigma}J^{\alpha\rho}). \end{aligned}$$

so that, in the end we have

$$i[J^{\alpha\beta}, J^{\rho\sigma}] = g^{\alpha\sigma} J^{\beta\rho} - g^{\alpha\rho} J^{\beta\sigma} - g^{\beta\sigma} J^{\alpha\rho} + g^{\beta\rho} J^{\alpha\sigma} \quad (4)$$

Part (b)

To obtain the irreducible representations of the Lorentz group we can decompose the generators in a way that they are elements of an Lie algebra by themselves. We can define the following generators:

$$\begin{cases} J_i = \frac{1}{2}\epsilon_{ijk}J^{jk} \\ K_i = J^{0i} \end{cases} \quad (5)$$

where J_i are the generators of three-dimensional rotations and K_i are the generators of boosts. Using the Lie algebra obtained in the previous part we can obtain the commutation relations between these generators:

First for the commutator between J_i and J_j we have

$$\begin{aligned} [J_i, J_j] &= \frac{1}{4}\epsilon_{ilm}\epsilon_{jnp}[J^{lm}, J^{np}] \\ &= \frac{i}{4}\epsilon_{ilm}\epsilon_{jnp}(g^{ln}J^{mp} - g^{lp}J^{mn} - g^{mn}J^{lp} + g^{mp}J^{ln}) \\ &= i\epsilon_{ijk}J_k \end{aligned} \quad (6)$$

↪ should have detailed more

for the commutator between J_i and K_j we have

$$\begin{aligned} [J_i, K_j] &= \frac{1}{2}\epsilon_{ilm}[J^{lm}, J^{0j}] \\ &= \frac{i}{2}\epsilon_{ilm}(g^{l0}J^{mj} - g^{lj}J^{m0} - g^{m0}J^{lj} + g^{mj}J^{l0}) \\ &= i\epsilon_{ijk}K_k \end{aligned} \quad (7)$$

and for the commutator between K_i and K_j we have

$$\begin{aligned} [K_i, K_j] &= [J^{0i}, J^{0j}] \\ &= i(g^{00}J^{ij} - g^{0j}J^{0i} - g^{0i}J^{j0} + g^{ij}J^{00}) \\ &= -i\epsilon_{ijk}J_k \end{aligned} \quad (8)$$

↪ again, more detail

in summary, the commutation relations between the generators of the Lorentz group can be written as

$$\begin{cases} [J_i, J_j] = i\epsilon_{ijk}J_k \\ [J_i, K_j] = i\epsilon_{ijk}K_k \\ [K_i, K_j] = -i\epsilon_{ijk}J_k \end{cases} \quad (9)$$

where we can see that the generator get mixed under the commutation, so that they do not form a Lie algebra by themselves, the idea here is to define new generators that are linear

combinations of J_i and K_i in such a way that they do not get mixed under the commutation. We can define the following generators:

$$\begin{cases} A_i = \frac{1}{2}(J_i + iK_i) \\ B_i = \frac{1}{2}(J_i - iK_i) \end{cases} \quad (10)$$

and using the commutation relations between J_i and K_i we can obtain the commutation relations between A_i and B_i :

$$\begin{cases} [A_i, A_j] = i\epsilon_{ijk}A_k \\ [B_i, B_j] = i\epsilon_{ijk}B_k \\ [A_i, B_j] = 0 \end{cases} \quad (11)$$

explicitly, the comutator between A_i and A_j is given by

$$\begin{aligned} [A_i, A_j] &= \frac{1}{4}([J_i, J_j] + i[J_i, K_j] + i[K_i, J_j] - [K_i, K_j]) \\ &= \frac{1}{4}(i\epsilon_{ijk}J_k + i^2\epsilon_{ijk}K_k + i^2\epsilon_{ijk}K_k + i\epsilon_{ijk}J_k) \\ &= i\epsilon_{ijk}A_k \end{aligned} \quad (12)$$

the comutator between B_i and B_j is given by

$$\begin{aligned} [B_i, B_j] &= \frac{1}{4}([J_i, J_j] - i[J_i, K_j] - i[K_i, J_j] - [K_i, K_j]) \\ &= \frac{1}{4}(i\epsilon_{ijk}J_k - i^2\epsilon_{ijk}K_k - i^2\epsilon_{ijk}K_k + i\epsilon_{ijk}J_k) \\ &= i\epsilon_{ijk}B_k \end{aligned} \quad (13)$$

and the comutator between A_i and B_j is given by

$$\begin{aligned} [A_i, B_j] &= \frac{1}{4}([J_i, J_j] + i[J_i, K_j] - i[K_i, J_j] + [K_i, K_j]) \\ &= \frac{1}{4}(i\epsilon_{ijk}J_k + i^2\epsilon_{ijk}K_k - i^2\epsilon_{ijk}K_k - i\epsilon_{ijk}J_k) \\ &= 0 \end{aligned} \quad (14)$$

so that, in the end we have indeed two generators that are independent e form and Lie algebra by themselves. Looking closely, we can also see that the commutation relations of both A_i and B_i are the same as the commutation relations of the generators of three-dimensional rotations, so that we can identify A_i and B_i as generators of two independent $SU(2)$ groups. If we index the representations of these two groups by j and j' respectively, we can label the irreducible representations of the Lorentz group by the pair (j, j') , where j and j' are half-integers. The representation $(0, 0)$ is the trivial representation for a scalar with no spin for example. Or we can have the representation $(\frac{1}{2}, 0)$ for a left-handed Weyl spinor and the representation $(0, \frac{1}{2})$ for a right-handed Weyl spinor, and the list goes on.

Problem 2. 1,5/2

Consider the four-vector electromagnetic potential

$$A^\mu = (A^0, \vec{A})$$

where ϕ ($\vec{A}$) is the scalar (vector) potential. In the absence of sources the electromagnetic Lagrangian density is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

- (a) Obtain the classical equations of motion.
- (b) Obtain the energy momentum tensor that is the conserved current when we consider the symmetry $x^\mu \rightarrow x'^\mu = x^\mu + a^\mu$ where a^μ is a constant

Solution**Part (a)**

To obtain the equations of motion we can use the Euler-Lagrange equations for vector fields:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} \right) - \frac{\partial \mathcal{L}}{\partial A_\nu} = 0. \quad (15)$$

calculating the derivatives by using the definition of $F_{\mu\nu}$ we have

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} &= -\frac{1}{4} \left(\frac{\partial F_{\alpha\beta}}{\partial(\partial_\mu A_\nu)} F^{\alpha\beta} + F_{\alpha\beta} \frac{\partial F^{\alpha\beta}}{\partial(\partial_\mu A_\nu)} \right) \\ &= -\frac{1}{4} \left(\frac{\partial(\partial_\alpha A_\beta - \partial_\beta A_\alpha)}{\partial(\partial_\mu A_\nu)} F^{\alpha\beta} + F_{\alpha\beta} \frac{\partial(\partial^\alpha A^\beta - \partial^\beta A^\alpha)}{\partial(\partial_\mu A_\nu)} \right) \\ &= -\frac{1}{4} ((\delta_\alpha^\mu \delta_\beta^\nu - \delta_\beta^\mu \delta_\alpha^\nu) F^{\alpha\beta} + F_{\alpha\beta} (\eta^{\mu\alpha} \eta^{\nu\beta} - \eta^{\mu\beta} \eta^{\nu\alpha})) \\ &= -\frac{1}{4} (F^{\mu\nu} - F^{\nu\mu} + F^{\mu\nu} - F^{\nu\mu}) = -F^{\mu\nu} \end{aligned} \quad (16)$$

where in the last step we used the antisymmetry of $F^{\mu\nu}$ that arises from its definition:

$$F^{\mu\nu} = (\partial_\mu A_\nu - \partial_\nu A_\mu) = -(\partial_\nu A_\mu - \partial_\mu A_\nu) = -F^{\nu\mu}. \quad (17)$$

Now, since $\mathcal{L}$ does not depend explicitly on A_ν we have $\frac{\partial \mathcal{L}}{\partial A_\nu} = 0$, so that the equations of motion are

$$\partial_\mu F^{\mu\nu} = 0. \quad (18)$$

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Part (b)

Since this is the first question related to the conserved current associated to the Noether's theorem, it's worth to make a small discussion about the procedure to obtain it. Considering an arbitrary transformation of the field (here the scalar one but throughout the problems it will be considered how it will transform specifically for each case) then the variation of the Lagrangian will be:

$$\delta\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\phi}\delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta(\partial_\mu\phi) \quad \text{only is true if } \phi \text{ does not transform, or if you are considering } \delta\phi = \phi'(x) - \phi(x) \quad (19)$$

with a change in the derivative through the product rule:

$$\delta\mathcal{L} = \left(\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \right) \delta\phi + \partial_\mu \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \delta\phi \right) \quad \text{otherwise there is an additional term} \quad (20)$$

if the equations of motion are satisfied, the first term of the sum will be zero and the variation of the Lagrangian density can be written as a total derivative:

$$\delta\mathcal{L} = \partial_\mu \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \delta\phi \right). \quad (21)$$

Since the transformation is a symmetry if the Lagrangian is invariant under it up to a total derivative, we can assume that $\delta\mathcal{L} = \partial_\mu K^\mu$ for some K^μ , so that the conserved current associated to the transformation is given by

$$\partial_\mu \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \delta\phi \right) = \partial_\mu K^\mu \Rightarrow \partial_\mu \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \delta\phi - K^\mu \right) = 0 \quad \text{or } \dots \quad (22)$$

and the conserved current associated to the transformation is

$$j^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \delta\phi - K^\mu. \quad (23)$$

Using Noether's theorem, the conserved current associated to spacetime translations is the energy-momentum tensor given by:

$$T^{\mu\nu} = \frac{\partial\mathcal{L}}{\partial(\partial_\mu A_\alpha)} \partial^\nu A_\alpha - g^{\mu\nu} \mathcal{L}. \quad \text{why all the discussion if you didn't even compute } \delta A_\alpha, \text{ neither } K^\mu. \quad (24)$$

now, using the results obtained in the previous part we have

$$T^{\mu\nu} = -F^{\mu\alpha} \partial^\nu A_\alpha + \frac{1}{4} g^{\mu\nu} F_{\alpha\beta} F^{\alpha\beta}. \quad \text{This is the right answer, but...} \quad (25)$$

even though this is already the desired conserved current, it is common to make some adjustments to the electromagnetic energy-momentum tensor. First, it's possible to rewrite the derivative of the potential term as:

$$-F^{\mu\alpha} \partial^\nu A_\alpha = -F^{\mu\alpha} (F_\alpha^\nu - \partial_\alpha A^\nu) = -F^{\mu\alpha} F_\alpha^\nu + F^{\mu\alpha} \partial_\alpha A^\nu \quad (26)$$

which allows us to write the energy-momentum tensor as

$$T^{\mu\nu} = -F^{\mu\alpha}F^\nu{}_\alpha + \frac{1}{4}g^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta} + F^{\mu\alpha}\partial_\alpha A^\nu. \quad (27)$$

Now, we can use the fact that the equations of motion need to be satisfied to rewrite the last term as

$$F^{\mu\alpha}\partial_\alpha A^\nu = \partial_\alpha(F^{\mu\alpha}A^\nu) - \underbrace{(\partial_\alpha F^{\mu\alpha})A^\nu}_{=0} = \partial_\alpha(F^{\mu\alpha}A^\nu) \quad (28)$$

and the tensor as

$$T^{\mu\nu} = -F^{\mu\alpha}F^\nu{}_\alpha + \frac{1}{4}g^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta} + \partial_\alpha(F^{\mu\alpha}A^\nu). \quad (29)$$

The advantage of this form is that the last term is a total derivative, so it does not contribute to the conserved charge associated to $T^{\mu\nu}$, which allow us to define a new energy-momentum tensor that is just as good as the previous one, but is symmetric in its indices:

$$\Theta^{\mu\nu} = -F^{\mu\alpha}F^\nu{}_\alpha + \frac{1}{4}g^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}. \quad (30)$$

To see clearly why this is possible, let's take the original tensor and write it's derivative

$$\partial_\mu T^{\mu\nu} = \partial_\mu \Theta^{\mu\nu} + \partial_\mu \partial_\alpha (F^{\mu\alpha}A^\nu) \quad (31)$$

the second term of the sum has two derivatives, which are comutative, being contracted with an antisymmetric tensor, so that

$$\begin{aligned} \partial_\mu \partial_\alpha (F^{\mu\alpha}A^\nu) &= \frac{1}{2}(\partial_\mu \partial_\alpha + \partial_\alpha \partial_\mu)(F^{\mu\alpha}A^\nu) \\ &= \frac{1}{2}[\partial_\mu \partial_\alpha (F^{\mu\alpha}A^\nu) + \underbrace{\partial_\alpha \partial_\mu (F^{\mu\alpha}A^\nu)}_{\mu \leftrightarrow \alpha}] \\ &= \frac{1}{2}[\partial_\mu \partial_\alpha (F^{\mu\alpha}A^\nu) + \partial_\mu \partial_\alpha (F^{\alpha\mu}A^\nu)] = \frac{1}{2}\partial_\mu \partial_\alpha (F^{\mu\alpha}A^\nu - F^{\alpha\mu}A^\nu) = 0 \end{aligned} \quad (32)$$

where for the last line it's used that the antisymmetry of $F^{\mu\alpha}$ implies that $F^{\alpha\mu} = -F^{\mu\alpha}$ and that the derivatives acting on A^ν are the same in both terms and will cancel each other.

To summarize, the conserved current associated to spacetime translations is the energy-momentum tensor, which can be written in a symmetric form as

$$\Theta^{\mu\nu} = -F^{\mu\alpha}F^\nu{}_\alpha + \frac{1}{4}g^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}. \quad (33)$$

and even though it's not strictly necessary to write it in this form, it is a common practice to do so. The property of being symmetric is useful to make calculations easier and to make the physical interpretation of the tensor more clear. Besides that, when working on other contexts, such as curved spacetime, the symmetry of the energy-momentum tensor is a necessary condition for it to be the source of spacetime curvature in Einstein's equations.

ok. Is nice this derivation
But the proper result
which was
you just cited.
You should have computed
 K^μ and δA^μ

Problem 3. 1,912

Consider a real scalar field ϕ whose Lagrangian density is

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{\lambda}{4!} \phi^4.$$

A scale transformation is defined as

$$x^\mu \rightarrow x'^\mu = e^\epsilon x^\mu, \quad \phi(x) \rightarrow \phi'(x') = e^{-d_\phi \epsilon} \phi(x),$$

where $d_\phi = 1$ is the dimension of the field ϕ . Is this transformation a symmetry? If so, what is the conserved current?

Solution

Using the form of the transformation given, we can first see under what conditions the Lagrangian density is invariant and if $d_\phi = 1$ is one of those conditions. We have

$$\begin{aligned} \mathcal{L}'(x') &= \frac{1}{2} \partial'_\mu \phi'(x') \partial'^\mu \phi'(x') - \frac{\lambda}{4!} (\phi'(x'))^4 \\ &= \frac{1}{2} e^{-2\epsilon(d_\phi+1)} \partial_\mu \phi(x) \partial^\mu \phi(x) - \frac{\lambda}{4!} e^{-4d_\phi \epsilon} (\phi(x))^4 \\ &= e^{-2\epsilon(d_\phi+1)} \frac{1}{2} \partial_\mu \phi(x) \partial^\mu \phi(x) - \frac{\lambda}{4!} e^{-4d_\phi \epsilon} (\phi(x))^4 \end{aligned} \quad (34)$$

where it was used that the derivative transforms as:

$$\partial'_\mu \phi' = \frac{\partial \phi'}{\partial x'^\mu} = \frac{\partial x^\nu}{\partial x'^\mu} \frac{\partial \phi'}{\partial x^\nu} = e^{-\epsilon} e^{-d_\phi \epsilon} \delta_\mu^\nu \partial_\nu \phi(x) = e^{-(d_\phi+1)\epsilon} \partial_\mu \phi(x). \quad (35)$$

Together with this, we can impose the condition that the action is invariant under the transformation, which means that

$$\int d^4 x' \mathcal{L}'(x') = \int d^4 x \mathcal{L}(x) \quad (36)$$

but since x transforms in such a way that

$$dx' = \left| \frac{\partial x'}{\partial x} \right| dx = e^\epsilon dx \quad (37)$$

we arrive at

$$\int e^{4\epsilon} d^4 x \left[e^{-2\epsilon(d_\phi+1)} \frac{1}{2} \partial_\mu \phi(x) \partial^\mu \phi(x) - \frac{\lambda}{4!} e^{-4d_\phi \epsilon} (\phi(x))^4 \right] = \int d^4 x \mathcal{L}(x) \quad (38)$$

which can be accomplished if

$$\begin{cases} 4\epsilon - 2\epsilon(d_\phi + 1) = 0 \\ 4\epsilon - 4d_\phi \epsilon = 0 \end{cases} \quad (39)$$

since this would imply in the exponential factors to be equal to 1 and the action would be invariant under the transformation. Solving the system of equations we have

$$\begin{cases} d_\phi = 1 \\ d_\phi = 1 \end{cases} \quad (40)$$

so that, indeed, the transformation is a symmetry of the theory. *ok*

Now, to obtain the conserved current associated to this symmetry we can use Noether's theorem and the transformation in the infinitesimal form. However, as we have just seen, even though the action is invariant, the Lagrangian density by itself isn't, so it will be necessary to write the variation of it in terms of the variations of coordinates, fields and derivatives of the fields. We have first for the field a transformation that can be written in two different ways:

$$\begin{cases} \phi'(x') = \phi'(x + \epsilon x) \approx \phi'(x) + \epsilon x^\mu \partial_\mu \phi'(x) \\ \phi'(x') = e^{-\epsilon} \phi(x) \approx \phi(x) - \epsilon \phi(x) \end{cases} \quad (41)$$

setting both expressions equal to each other we have

$$\delta\phi(x) = \phi'(x) - \phi(x) = -\epsilon(1 + x^\mu \partial_\mu)\phi(x) \quad \text{ok} \quad (42)$$

using that in first order in ϵ we can write $\partial_\mu \phi'(x) \approx \partial_\mu \phi(x)$, since this whole term is already of order ϵ , expanding again the field would just contribute to a second order term, which we can neglect. Now, for the variation of the derivative of the field we have

$$\begin{aligned} \delta(\partial_\mu \phi) &= \partial_\mu [-\epsilon(1 + x^\nu \partial_\nu)\phi(x)] \\ &= -\epsilon [\partial_\mu \phi(x) + \partial_\mu (x^\nu \partial_\nu \phi(x))] \\ &= -\epsilon [\partial_\mu \phi(x) + \delta_\mu^\nu \partial_\nu \phi(x) + x^\nu \partial_\mu \partial_\nu \phi(x)] \\ &= -\epsilon [2\partial_\mu \phi(x) + x^\nu \partial_\mu \partial_\nu \phi(x)] \end{aligned} \quad (43)$$

Finally, for the variation of the Lagrangian density we have

$$\begin{aligned} \delta\mathcal{L} &= \frac{\partial\mathcal{L}}{\partial\phi} \delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu \phi)} \delta(\partial_\mu \phi) \\ &= \left(-\frac{\lambda}{3!} \phi^3\right) [-\epsilon(1 + x^\mu \partial_\mu)\phi(x)] + (\partial^\mu \phi(x)) [-\epsilon(2\partial_\mu \phi(x) + x^\nu \partial_\mu \partial_\nu \phi(x))] \\ &= -\epsilon \left[-\frac{\lambda}{3!} \phi^4 \oplus \frac{\lambda}{3!} x^\mu \partial_\mu \phi(x) + 2\partial^\mu \phi(x) \partial_\mu \phi(x) + x^\nu \partial^\mu \phi \partial_\mu \partial_\nu \phi \right] \quad (44) \\ &= -\epsilon \left[-4 \left(\frac{\lambda}{4!} \phi^4\right) \ominus x^\nu \partial_\nu \left(\frac{\lambda}{4!} \phi^4\right) + 4 \left(\frac{1}{2} \partial^\mu \phi \partial_\mu \phi\right) + x^\nu \partial_\nu \left(\frac{1}{2} \partial^\mu \phi \partial_\mu \phi\right) \right] \\ &= -\epsilon [-4\mathcal{L} \ominus x^\nu \partial_\nu \mathcal{L}] = -\epsilon \partial_\mu (x^\mu \mathcal{L}) \rightarrow \text{correct sign overall} \\ &\quad \text{wrong sign} \end{aligned}$$

so, it's possible to write the variation of the Lagrangian density as a total derivative:

$$\delta\mathcal{L} = -\epsilon \partial_\mu (x^\mu \mathcal{L}). \quad \text{ok} \dots \quad (45)$$

Inserting everything into the conserved current equation give by the theorem:

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta \phi \ominus x^\mu \mathcal{L} = -\epsilon \partial^\mu \phi(x) (1 + x^\nu \partial_\nu) \phi + \epsilon x^\mu \mathcal{L} \quad (46)$$

$\rightarrow$ missing an ϵ , wrong sign
 $\hookrightarrow$ ok

which will give the conserved current, up to the factor ϵ , associated to the scale transformation:

$$j^\mu = \phi \partial^\mu \phi + x^\nu \partial^\mu \phi \partial_\nu \phi(x) - x^\mu \mathcal{L}. \quad (47)$$

ok ...

Some modifications can be made to the current, such as writing the second term as something related to the energy-momentum tensor in order to make the physical interpretation of the current more clear:

$$j^\mu = \phi \partial^\mu \phi + x^\nu \underbrace{(\partial^\mu \phi \partial_\nu \phi(x) - \delta^\mu_\nu \mathcal{L})}_{T^\mu_\nu}. \quad (48)$$

$$j^\mu = \phi \partial^\mu \phi + x^\nu T^\mu_\nu. \quad (49)$$

ok

We arrived at the current written in terms of the energy-momentum tensor and the field itself. For this type of transformation, the interpretation of the current is not as straightforward as for other symmetries, but can be thought as a current that measures how the field and its energy-momentum distribution change around the spacetime.

Problem 4. 2/2

Consider a Lorentz transformation

$$x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu}.$$

An infinitesimal Lorentz transformation takes the form

$$x'^{\mu} = x^{\mu} + \delta\omega^{\mu}_{\nu} x^{\nu}.$$

- (a) Show that $\delta\omega_{\mu\nu} = -\delta\omega_{\nu\mu}$.
- (b) For a rotation, what is the form of $\delta\omega_{\mu\nu}$?
- (c) For a scalar field $\phi(x)$ obtain the conserved current associated to a Lorentz transformation.

Solution**Part (a)**

Writing the Lorentz transformation in the infinitesimal form we have

$$\Lambda^{\mu}_{\nu} = \delta^{\mu}_{\nu} + \delta\omega^{\mu}_{\nu} \quad (50)$$

together with the condition that defines the Lorentz group:

$$\Lambda^{\mu}_{\alpha} \Lambda^{\nu}_{\beta} g^{\alpha\beta} = g^{\mu\nu} \quad (51)$$

we can restrict the form of $\delta\omega^{\mu}_{\nu}$. Putting them together we have

$$\begin{aligned} (\delta^{\mu}_{\alpha} + \delta\omega^{\mu}_{\alpha})(\delta^{\nu}_{\beta} + \delta\omega^{\nu}_{\beta})g^{\alpha\beta} &= g^{\mu\nu} \\ g^{\mu\nu} + \delta\omega^{\mu}_{\alpha}g^{\alpha\nu} + \delta\omega^{\nu}_{\beta}g^{\mu\beta} + O(\delta\omega^2) &= g^{\mu\nu} \end{aligned} \quad (52)$$

and neglecting the second order term

$$\begin{aligned} \delta\omega^{\mu}_{\alpha}g^{\alpha\nu} + \delta\omega^{\nu}_{\beta}g^{\mu\beta} &= 0 \\ \Rightarrow \delta\omega^{\mu\nu} + \delta\omega^{\nu\mu} &= 0 \\ \Rightarrow \delta\omega_{\mu\nu} &= -\delta\omega_{\nu\mu} \end{aligned} \quad (53) \quad \text{or}$$

where we reach the desired.

Part (b)

For the general rotation by an angle $\delta\theta$ around the axis $\hat{n}$, the transformation can be written as

$$\begin{aligned} \delta x &= \delta\theta(\hat{n} \times \mathbf{x}) \\ \Rightarrow \delta x_i &= \epsilon^{ijk} \delta\theta_j x_k \end{aligned} \quad (54)$$

which can be compared to the general form of the infinitesimal Lorentz transformation $\delta x^\mu = \delta\omega^\mu{}_\nu x^\nu$ in the case that only the spatial components are considered to obtain the form of $\delta\omega_{\mu\nu}$ for a rotation:

$$\epsilon^{ijk}\delta\theta_j x_k = \delta\omega^{ik} x_k \Rightarrow \delta\omega^{ik} = \epsilon^{ijk}\delta\theta_j \quad (55)$$

so, writing the antisymmetric matrix $\delta\omega^{\mu\nu}$ for a general rotation, we have

$$\delta\omega^{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -\delta\theta_3 & \delta\theta_2 \\ 0 & \delta\theta_3 & 0 & -\delta\theta_1 \\ 0 & -\delta\theta_2 & \delta\theta_1 & 0 \end{pmatrix} \quad \begin{array}{l} \text{named to comment} \\ \text{on } x^\mu \text{ not} \\ \text{being transformed} \\ \Rightarrow \omega_{0\mu} = 0. \end{array} \quad (56)$$

Part (c)

For the scalar field, we can start by writing the variation of the field. After, we proceed to the variation of the Lagrangian density and then we can use Noether's theorem to obtain the conserved current. We have first for the field

$$\begin{aligned} \phi'(x') &= \phi(\Lambda^{-1}x) = \phi(x - \delta\omega x) \approx \phi(x) - \delta\omega^\mu{}_\nu x^\nu \partial_\mu \phi(x) \\ &\Rightarrow \delta\phi(x) = -\delta\omega^\mu{}_\nu x^\nu \partial_\mu \phi(x). \end{aligned} \quad (57)$$

good.

We can now use the fact that the Lagrangian density is a scalar under Lorentz transformations and so it must transform in the same way as the field, which means that

$$\begin{aligned} \mathcal{L}'(x') &= \mathcal{L}(\Lambda^{-1}x) = \mathcal{L}(x - \delta\omega x) \approx \mathcal{L}(x) - \delta\omega^\mu{}_\nu x^\nu \partial_\mu \mathcal{L}(x) \\ &= \mathcal{L}(x) + \delta\mathcal{L} \\ &\Rightarrow \delta\mathcal{L} = -\delta\omega^\mu{}_\nu x^\nu \partial_\mu \mathcal{L}(x) \end{aligned} \quad (58)$$

nice. This was the expected to be done on (2)

however, for the theorem it's necessary to write the variation of the Lagrangian density as a total derivative. Starting from

$$\begin{aligned} \partial_\nu(\delta x^\nu \mathcal{L}) &= \partial_\nu(\delta x^\nu) \mathcal{L} + \delta x^\nu (\partial_\nu \mathcal{L}) \\ &= \partial_\nu(\delta\omega^\nu{}_\alpha x^\alpha) \mathcal{L} + \underbrace{\delta\omega^\nu{}_\alpha x^\alpha (\partial_\nu \mathcal{L})}_{-\delta\mathcal{L}} \end{aligned} \quad (59)$$

using the fact that $\delta\omega^\nu{}_\alpha$ is a global parameter, so it does not depend on the coordinates, we have

$$\partial_\nu(\delta x^\nu \mathcal{L}) = \delta\omega^\nu{}_\alpha \delta^\alpha{}_\nu \mathcal{L} - \delta\mathcal{L} = \delta\omega^\nu{}_\nu \mathcal{L} - \delta\mathcal{L} = -\delta\mathcal{L} \quad (60)$$

which can be seen as true using the fact that $\delta\omega^\nu{}_\alpha$ is antisymmetric, so that its trace must be zero.

Putting together $\delta\phi$ and $\delta\mathcal{L}$ in the equation for the conserved current:

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta\phi + \delta x^\mu \mathcal{L} = -\delta\omega^\alpha{}_\nu x^\nu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\alpha \phi - \delta^\mu{}_\alpha \mathcal{L} \right) \quad (61)$$

wrong sign
 $\delta x^\mu = \delta\omega^\mu{}_\nu x^\nu$
so maximally here is correct

once again we can identify the term in parentheses as the energy-momentum tensor, so that we can write the current as

$$j^\mu = -\delta\omega_{\alpha\nu} x^\nu T^{\mu\alpha}. \quad (62)$$

The form obtain is already the desired, up to the factor $\delta\omega_{\alpha\nu}$, but it's common to write the current in terms of another tensor that do not depend on the arbitrary parameter $\delta\omega_{\alpha\nu}$:

$$\begin{aligned}
 j^\mu &= -\delta\omega_{\alpha\nu} x^\nu T^{\mu\alpha} \\
 &= -\frac{1}{2}(\omega_{\alpha\nu} - \omega_{\nu\alpha}) x^\nu T^{\mu\alpha} \\
 &= -\frac{1}{2}\omega_{\rho\sigma}(\delta^\rho_\alpha \delta^\sigma_\nu - \delta^\sigma_\alpha \delta^\rho_\nu) x^\nu T^{\mu\alpha} \\
 &= -\frac{1}{2}\omega_{\rho\sigma}(x^\sigma T^{\mu\rho} - x^\rho T^{\mu\sigma}) = -\frac{1}{2}\omega_{\rho\sigma} M^{\mu\rho\sigma}. \quad \text{ovs !}
 \end{aligned} \tag{63}$$

The tensor $M^{\mu\rho\sigma} = x^\sigma T^{\mu\rho} - x^\rho T^{\mu\sigma}$ is called the angular momentum tensor and it's conservation $\partial_\mu M^{\mu\rho\sigma} = 0$ gives rise to 6 different currents, one for each independent component of the antisymmetric tensor $\omega_{\rho\sigma}$. When the index are spatial, the three conserved charges give the total angular momentum of the system:

$$J^i = \frac{1}{2}\epsilon^{ijk} \int d^3x M^{0jk} = \int d^3x (x^j T^{0k} - x^k T^{0j}) \tag{64}$$

as for the components with time index, the conserved charges are related to the boost of the system and are given by

$$K^i = \int d^3x (x^i T^{00} - x^0 T^{0i}). \tag{65}$$

the temporal conservation of this charge can help us to understand how the system behaves:

$$\frac{dK^i}{dt} = \int d^3x \partial_0 (x^i T^{00}) - \underbrace{[(\partial_0 x^0) T^{0i} + x^0 (\partial_0 T^{0i})]}_1 = 0 \tag{66}$$

and using the linear moment of the field

$$P^i = \int d^3x T^{0i} \tag{67}$$

we have

$$\frac{dK^i}{dt} = -\frac{d}{dt} \left(\int d^3x x^i T^{00} \right) + P^i + t \frac{dP^i}{dt} = 0 \tag{68}$$

since the linear momentum is also conserved, we have $\frac{dP^i}{dt} = 0$, so that

$$\frac{d}{dt} \left(\int d^3x x^i T^{00} \right) = P^i = \text{constant} \tag{69}$$

which means that the center of energy of the system moves with a constant velocity.

nice

Problem 5. 1,9/2

Consider the vector field A^μ associated to the electromagnetic field. Knowing that

$$A'^\mu(x') = \Lambda^\mu_\nu A^\nu(x),$$

obtain the conserved quantities due to the Lorentz invariance of the theory. Use the Lagrangian density given in the first problem.

Solution

For this case, now the field is itself a vector, so it transforms under Lorentz transformations as one does. Besides that, the spacetime coordinates also transform, so to first calculate the variation of the field we need to take into account both transformations. The total transformation of the field can be written as

$$A'^\mu(x') = \Lambda^\mu_\nu A^\nu(x) \approx (\delta^\mu_\nu + \delta\omega^\mu_\nu) A^\nu(x) = A^\mu(x) + \delta\omega^\mu_\nu A^\nu(x) \quad (70)$$

and since the variation is taken to be calculated at the same point, we can write

$$\begin{aligned} \delta A^\mu &= A'^\mu(x) - A^\mu(x) = A'^\mu(x' - \delta x') - A^\mu(x) \\ &= A'^\mu(x') - \delta x'^\nu \partial_\nu A'^\mu(x') - A^\mu(x) \\ &= \delta\omega^\mu_\nu A^\nu(x) - \delta x^\nu \partial_\nu A'^\mu(x') \end{aligned} \quad (71)$$

and for the derivative present on the second term, using the fact that ω is a global parameter::

$$\partial_\nu A'^\mu(x') = \partial_\nu [(\delta^\mu_\alpha + \delta\omega^\mu_\alpha) A^\alpha(x)] = (\delta^\mu_\alpha + \delta\omega^\mu_\alpha) \partial_\nu A^\alpha(x) \quad (72)$$

inserting in the variation we are left with

$$\begin{aligned} \delta A^\mu &= \delta\omega^\mu_\nu A^\nu(x) - \delta x^\nu (\delta^\mu_\alpha + \delta\omega^\mu_\alpha) \partial_\nu A^\alpha(x) \\ &= \delta\omega^\mu_\nu A^\nu(x) - \delta x^\nu \partial_\nu A^\mu(x) - \underbrace{\delta x^\nu \delta\omega^\mu_\alpha}_{\mathcal{O}(\omega^2)} \partial_\nu A^\alpha(x) \end{aligned} \quad (73)$$

that gives the variation of the field as

$$\begin{aligned} \delta A^\mu &= \delta\omega^\mu_\nu A^\nu(x) - \delta x^\nu \partial_\nu A^\mu(x) \\ &= \delta\omega^\mu_\nu A^\nu(x) - \delta\omega^\nu_\alpha x^\alpha \partial_\nu A^\mu(x) \\ &= \delta\omega^\nu_\rho [-x^\rho \partial_\nu A^\mu(x) \ominus \delta^\mu_\nu A^\rho(x)] \end{aligned} \quad (74)$$

Now, for the variation of the Lagrangian density we can use again the fact that it is a scalar under Lorentz transformations, so it transforms as in (60). Putting everything together in the equation for the conserved current we have

$$\begin{aligned} j^\mu &= \frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\alpha)} \delta A_\alpha \ominus \delta x^\mu \mathcal{L} \\ &= -\delta\omega_{\nu\rho} \left[x^\rho \frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\alpha)} \partial^\nu A_\alpha \oplus \frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\alpha)} \eta^\nu_\alpha A^\rho(x) \oplus x^\rho \delta^{\mu\nu} \mathcal{L} \right] \end{aligned} \quad (75)$$

as we also have seen early, the derivative of the Lagrangian density with respect to the derivative of the field is given by (16), so that we can write the current as

$$j^\mu = -\delta\omega_{\nu\rho} [x^\rho (-F^{\mu\alpha} \partial^\nu A_\alpha) \oplus (-F^{\mu\nu} A^\rho(x)) \oplus x^\rho \delta^{\mu\nu} \mathcal{L}]$$

$$= -\delta\omega_{\nu\rho} [-x^\rho F^{\mu\alpha} \partial^\nu A_\alpha \ominus F^{\mu\nu} A^\rho(x) \oplus x^\rho \delta^{\mu\nu} \mathcal{L}]$$
(76)

once again identifying the term of the energy-momentum tensor as in part b of problem 1:

$$j^\mu = -\delta\omega_{\nu\rho} [x^\rho T^{\mu\nu} \ominus F^{\mu\nu} A^\rho]$$
(77)

For making the physical interpretation of the current more clear, we can write in a similar manner as in the previous problem, the current in terms of another tensor that does not depend on the arbitrary parameter $\delta\omega_{\nu\rho}$:

$$j^\mu = -\frac{1}{2} \delta\omega_{\alpha\beta} (\delta^\alpha_\nu \delta^\beta_\rho - \delta^\beta_\nu \delta^\alpha_\rho) [-x^\rho T^{\mu\nu} - F^{\mu\nu} A^\rho]$$

$$= -\frac{1}{2} \delta\omega_{\alpha\beta} [(x^\alpha T^{\mu\beta} - x^\beta T^{\mu\alpha}) + (F^{\mu\beta} A^\alpha - F^{\mu\alpha} A^\beta)] = -\frac{1}{2} \omega_{\alpha\beta} M^{\mu\alpha\beta}$$
(78)

where we defined the angular momentum tensor for the electromagnetic field as

$$M^{\mu\alpha\beta} = (x^\alpha T^{\mu\beta} - x^\beta T^{\mu\alpha}) + (F^{\mu\beta} A^\alpha - F^{\mu\alpha} A^\beta).$$
(79)

Comparing this result with what was obtained for the scalar field, we can see that the current has an extra term, which is related to the fact that the field itself transforms under Lorentz transformations. The first term is the orbital angular momentum tensor, as before. The second one is the spin angular momentum tensor, which is related to the intrinsic angular momentum of the field itself that arises from the fact that the field is a vector field and has also a transformation under Lorentz. The conservation of the current $\partial_\mu j^\mu = 0$ implies in the conservation of the angular momentum tensor $\partial_\mu M^{\mu\alpha\beta} = 0$, corresponding to the total angular momentum of the system, which is the sum of the orbital and spin angular momentum.